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Prof. Ing. Jan Flusser, DrSc.

Digitální zpracování obrazu

Lecture 1

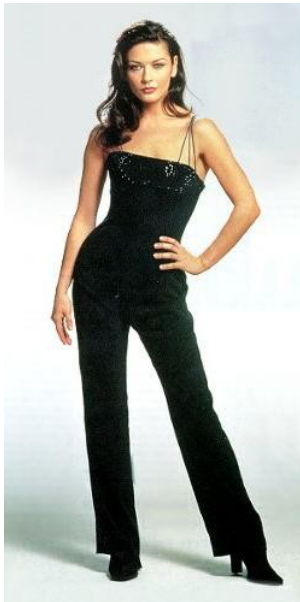
Důležité informace

- Rozsah: ZS, 3+0, Zk
- <http://zoi.utia.cz/teaching>

Navazující předměty

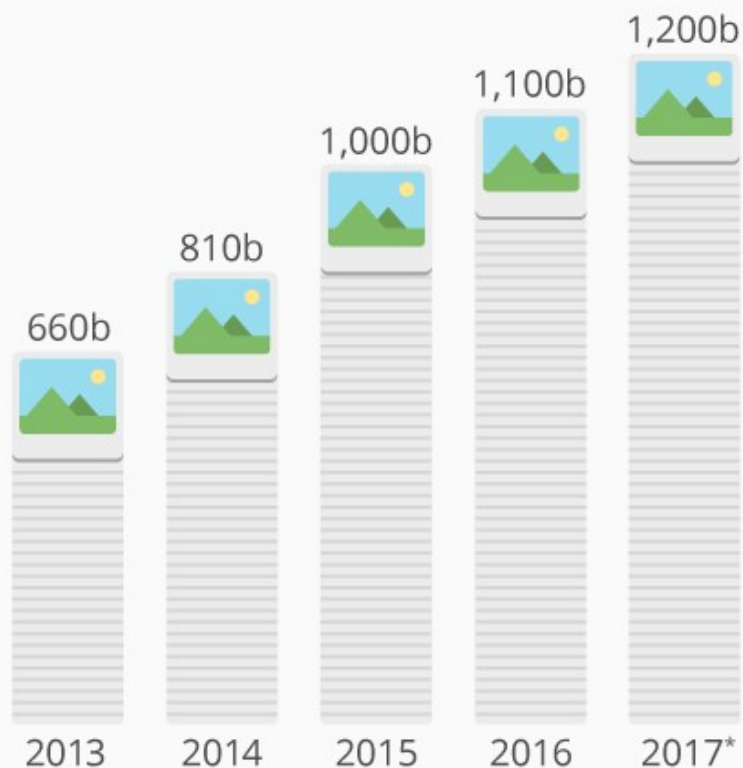
- **NPGR013 (J. Flusser, B. Zitová)**
- **NPGR022 (J. Flusser, B. Zitová)**
- **NPGR029 (F. Šroubek, J. Flusser)**
- **NPGR032 (A. Novozámský, B. Zitová)**

Digitálních fotografií je všude plno ...

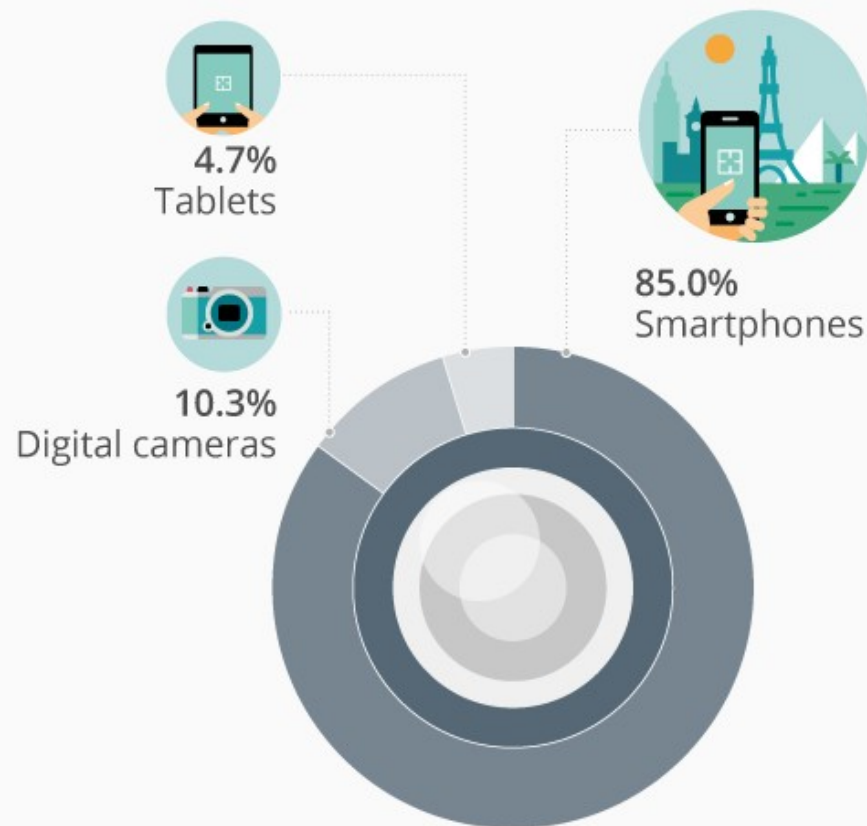


Smartphones Cause Photography Boom

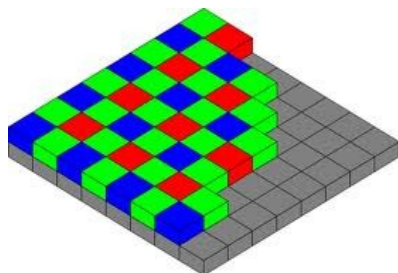
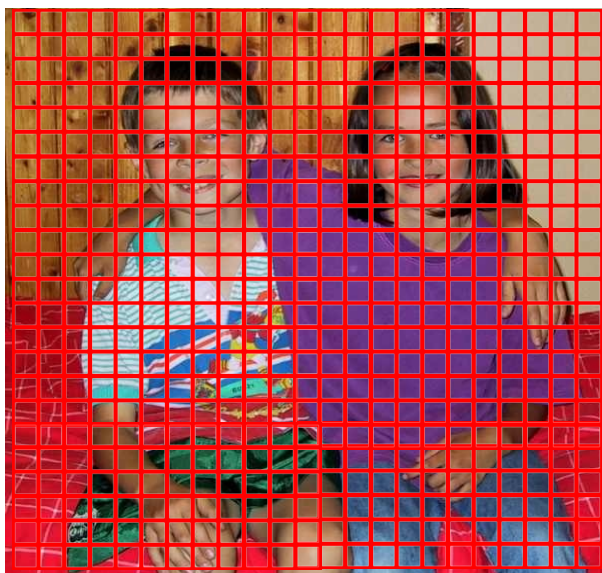
Number of digital photos taken worldwide*



Devices used in 2017



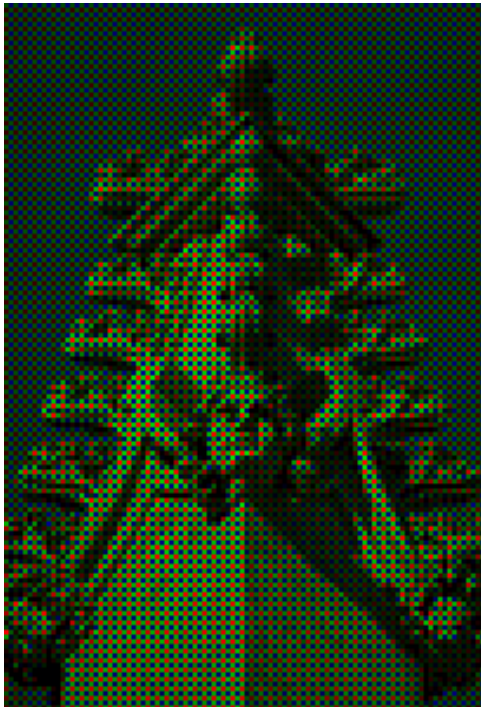
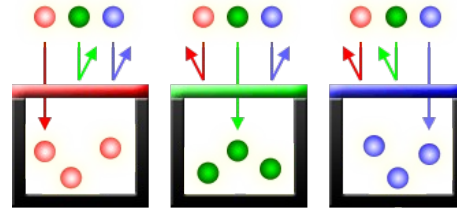
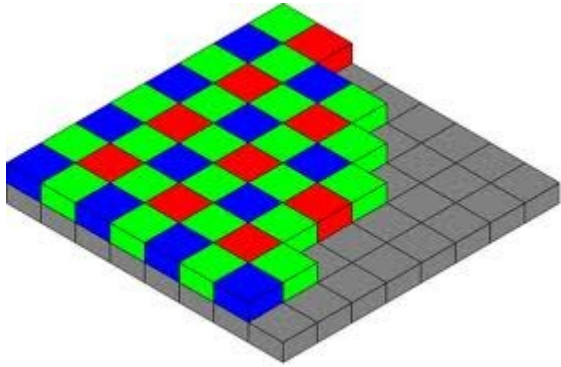
Jak vzniknou?



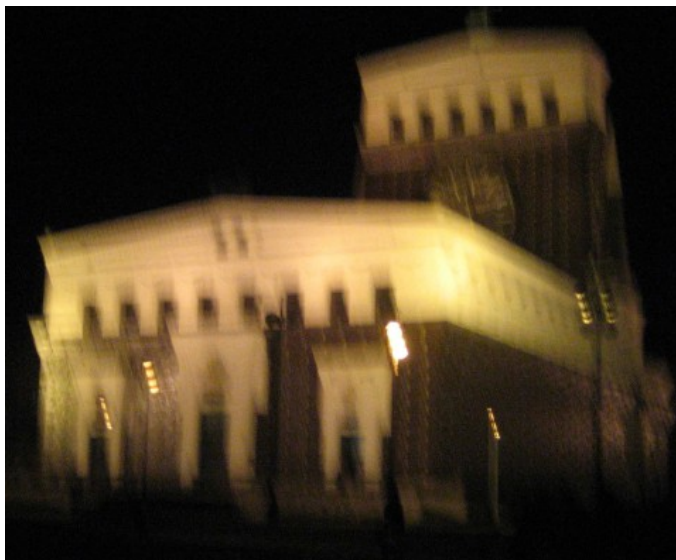
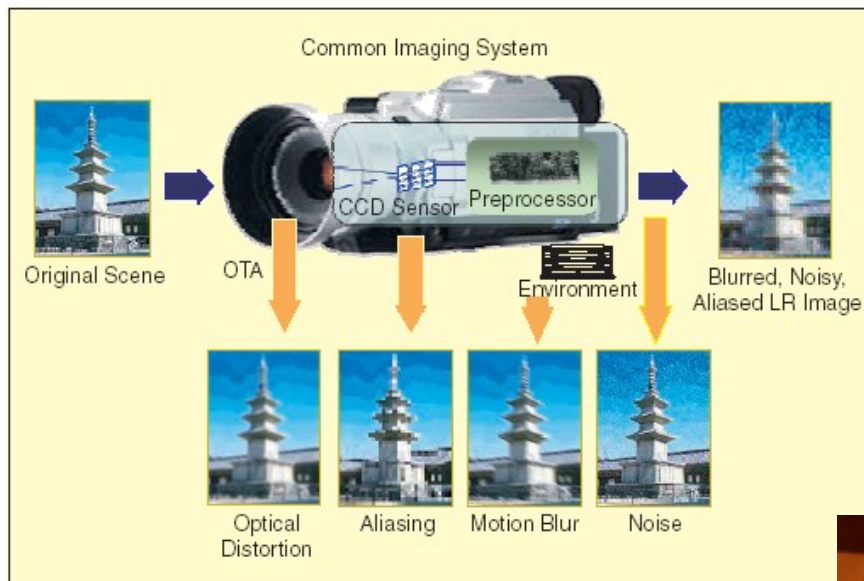
$$\begin{bmatrix} 16 & 11 & 10 & 16 & 24 & 40 & 51 & 61 \\ 12 & 12 & 14 & 19 & 26 & 58 & 60 & 55 \\ 14 & 13 & 16 & 24 & 40 & 57 & 69 & 56 \\ 14 & 17 & 22 & 29 & 51 & 87 & 80 & 62 \\ 18 & 22 & 37 & 56 & 68 & 109 & 103 & 77 \\ 24 & 35 & 55 & 64 & 81 & 104 & 113 & 92 \\ 49 & 64 & 78 & 87 & 103 & 121 & 120 & 101 \\ 72 & 92 & 95 & 98 & 112 & 100 & 103 & 99 \end{bmatrix}$$


original downsampled

Jak se tam dostanou barvy?

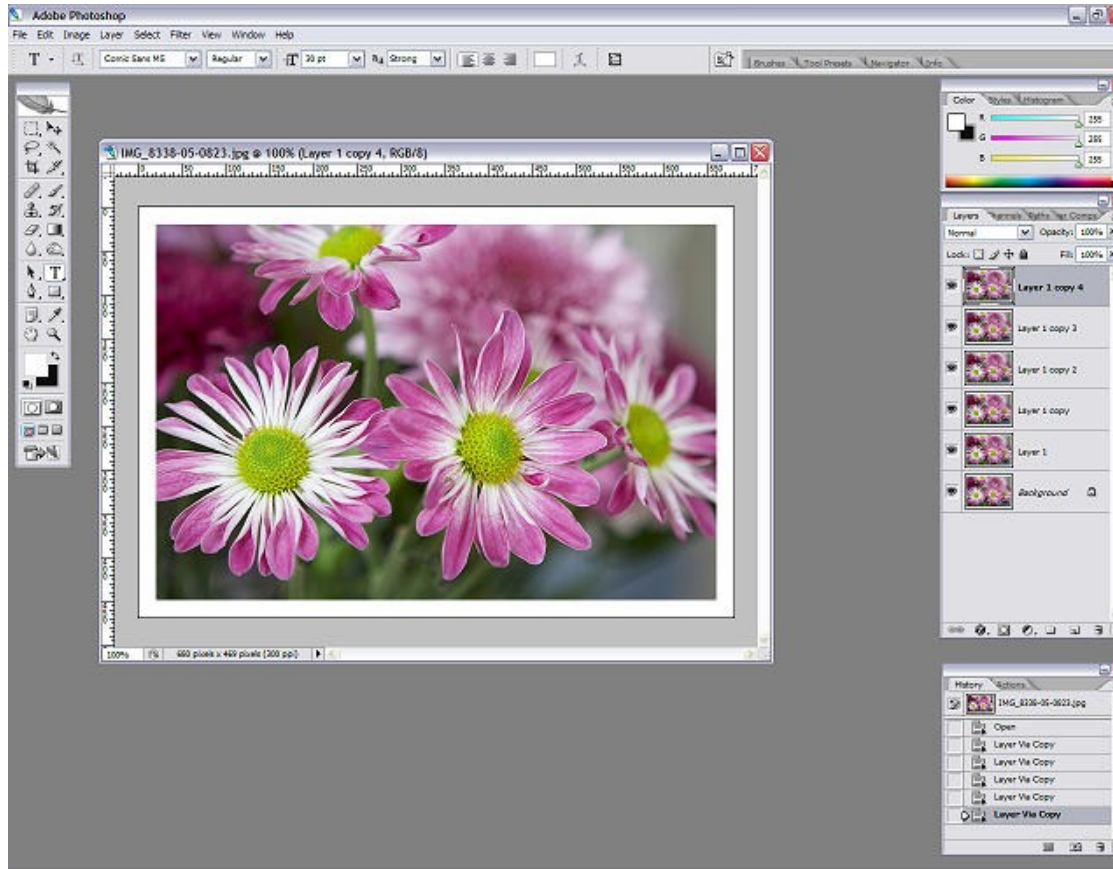


Fotografie nebývají ideální



Jak vylepšit nekvalitní foto?

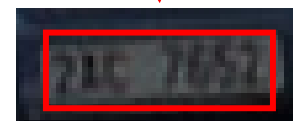
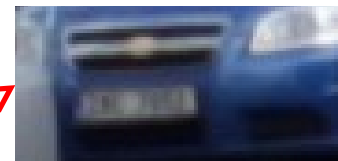
Všichni známe Photoshop ...



... na jednoduché úpravy stačí



Kde Photoshop nestačí



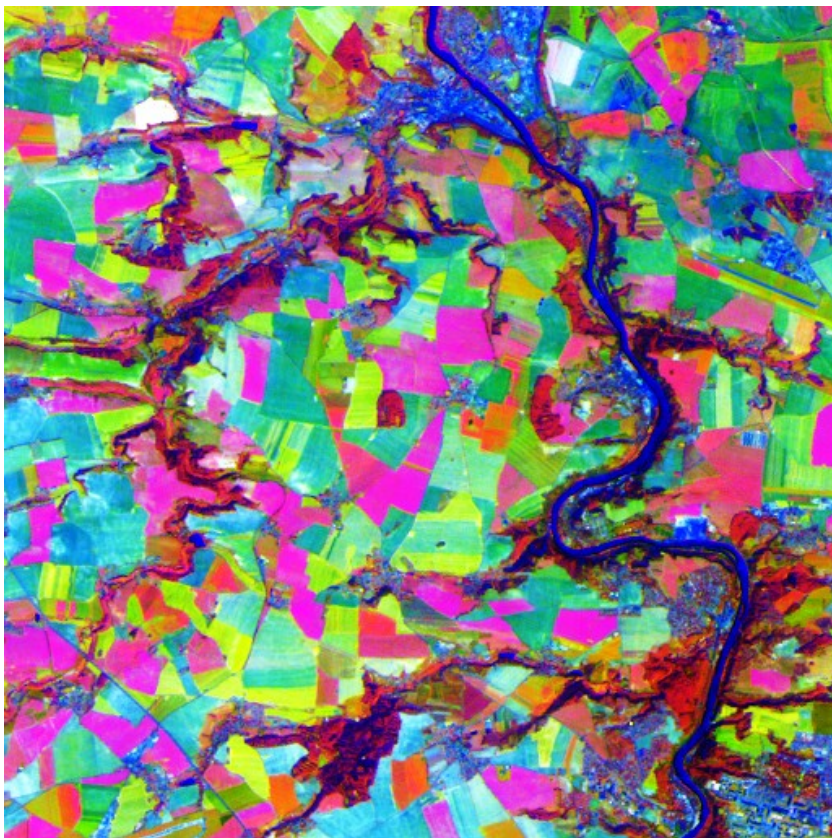
$(F_1, F_2, \dots, F_n)$

2AC 7652

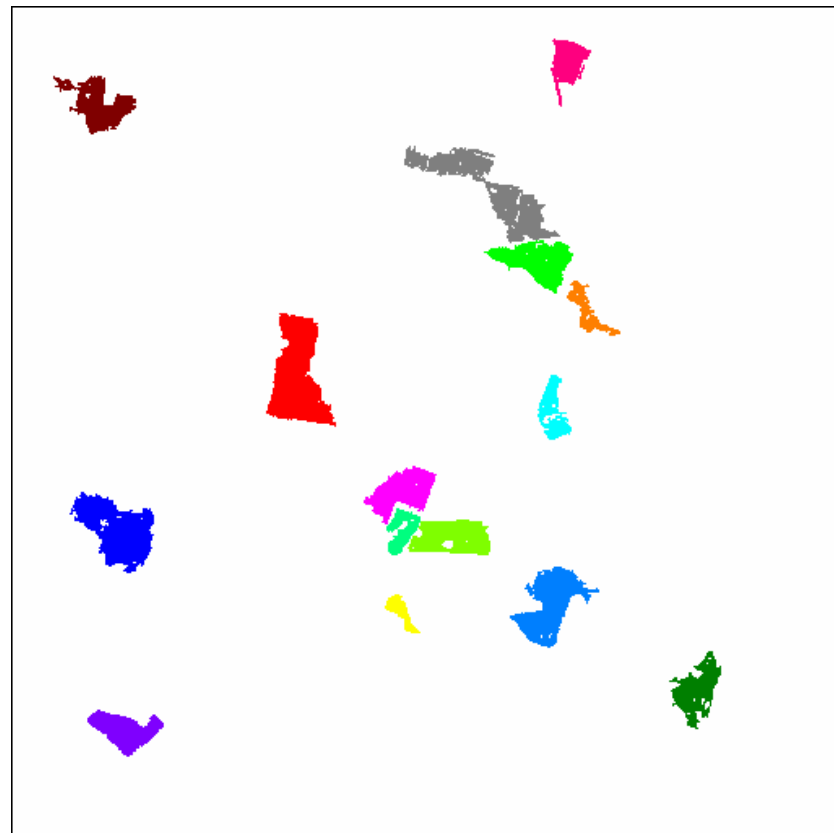
Jak odstranit třes ruky?



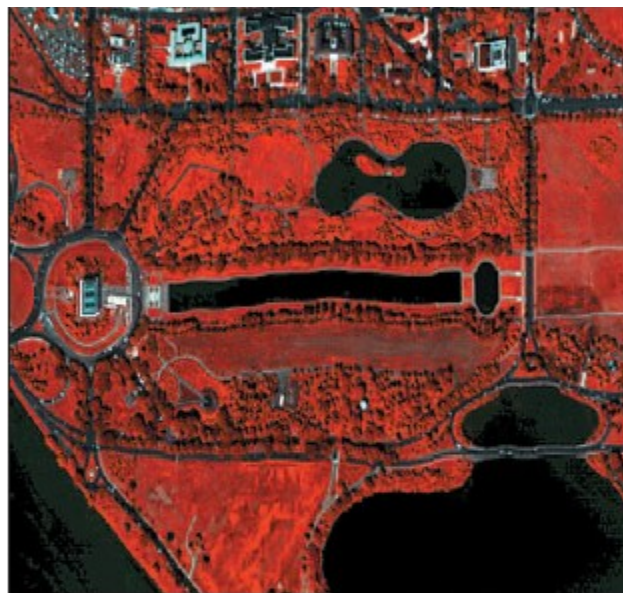
Segmentace



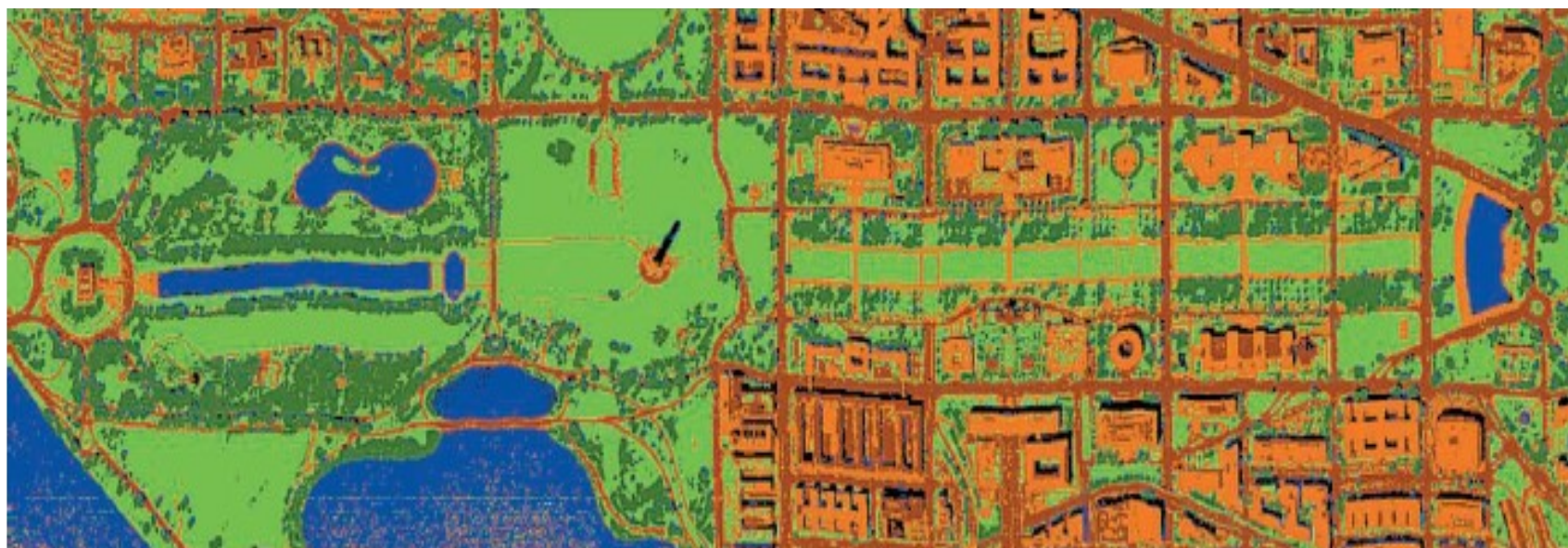
Originál



Detekované objekty



- Groups
- background
 - Roofs
 - Road
 - Grass
 - Trees
 - Trail
 - Water
 - Shadow

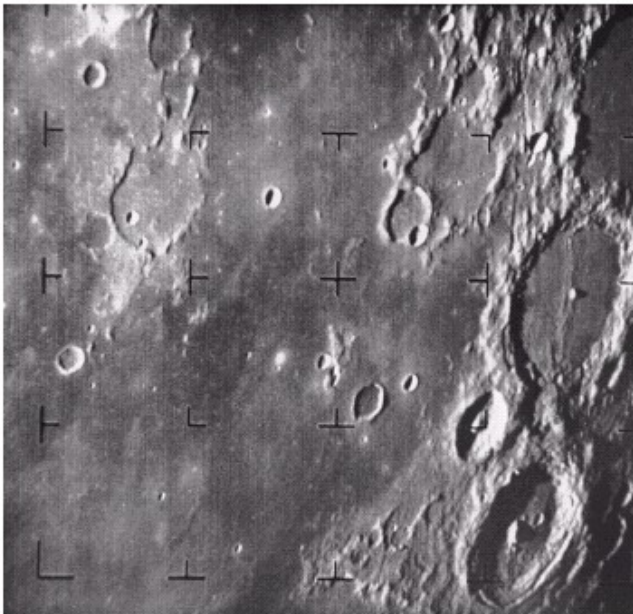


Historie



1921

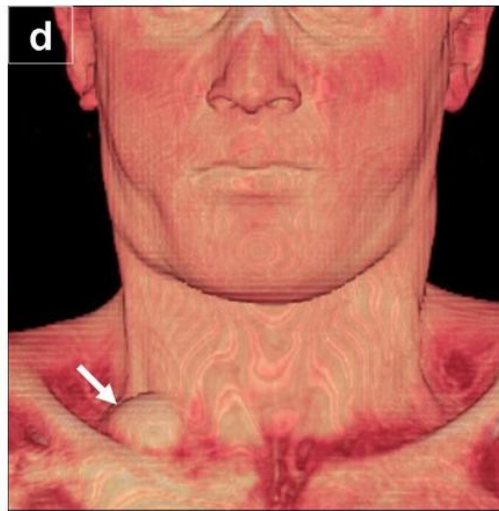
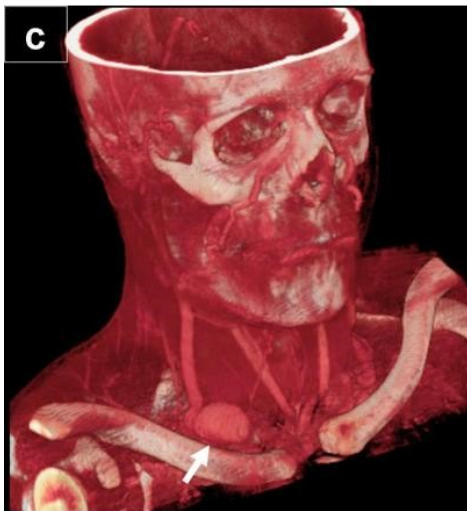
1929



1964

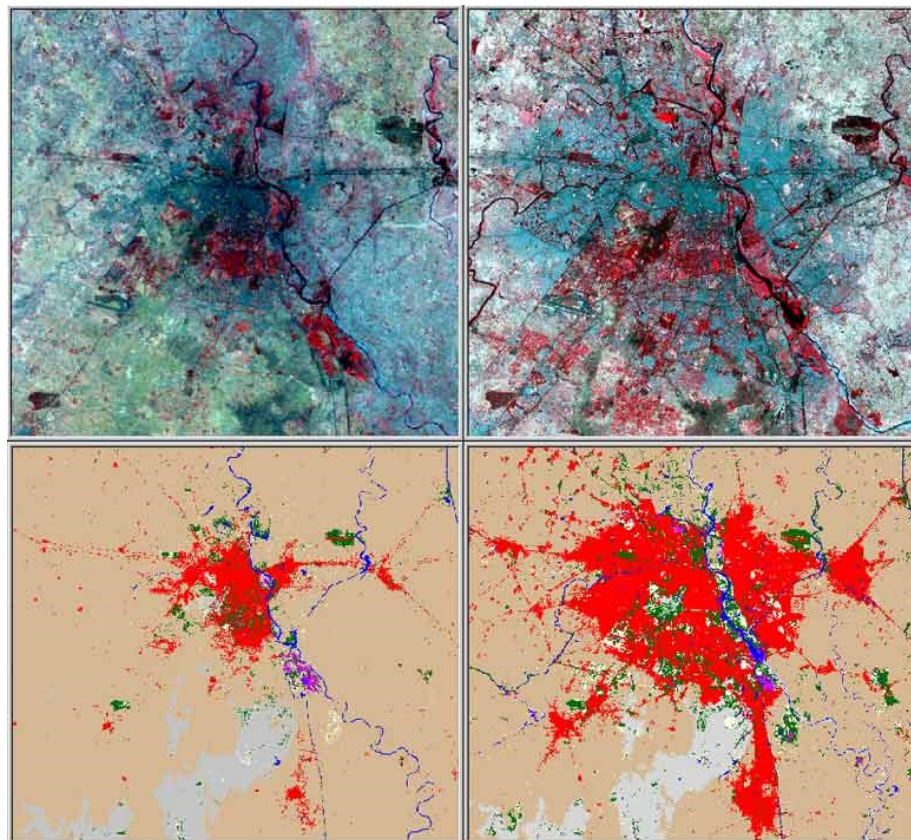
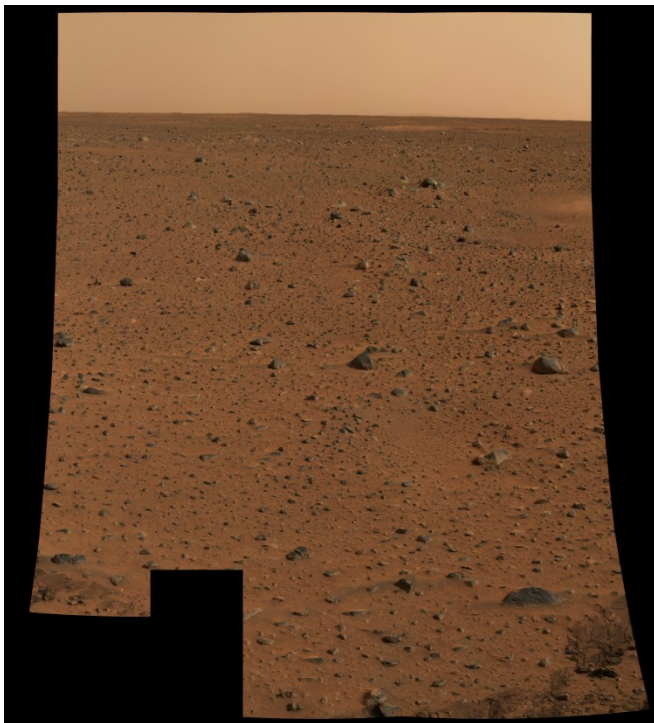
Aplikační oblasti

- medicína



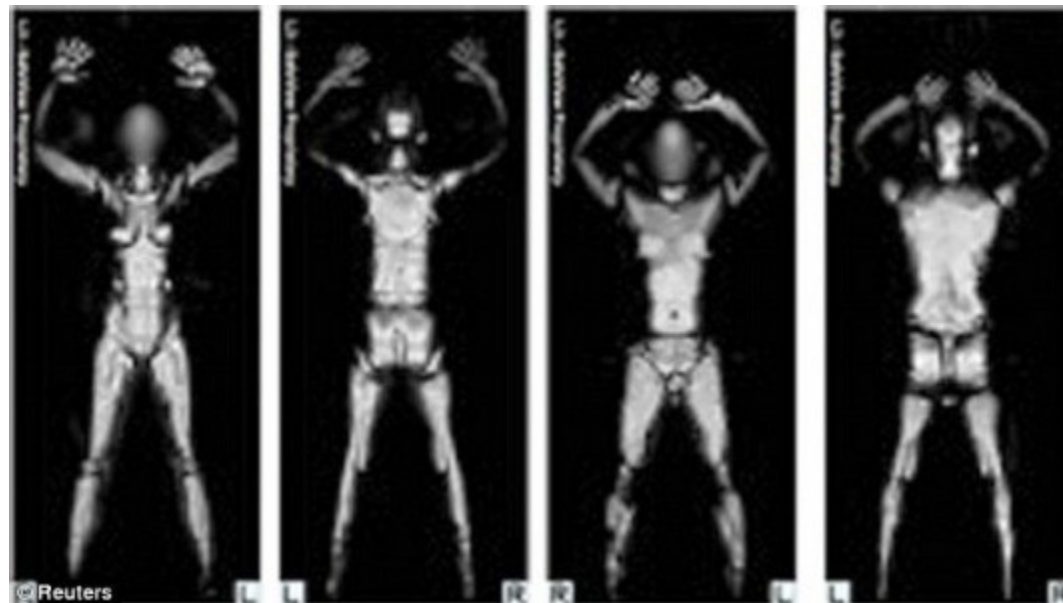
Aplikační oblasti

- Dálkový průzkum



Aplikační oblasti

- Bezpečnost, doprava

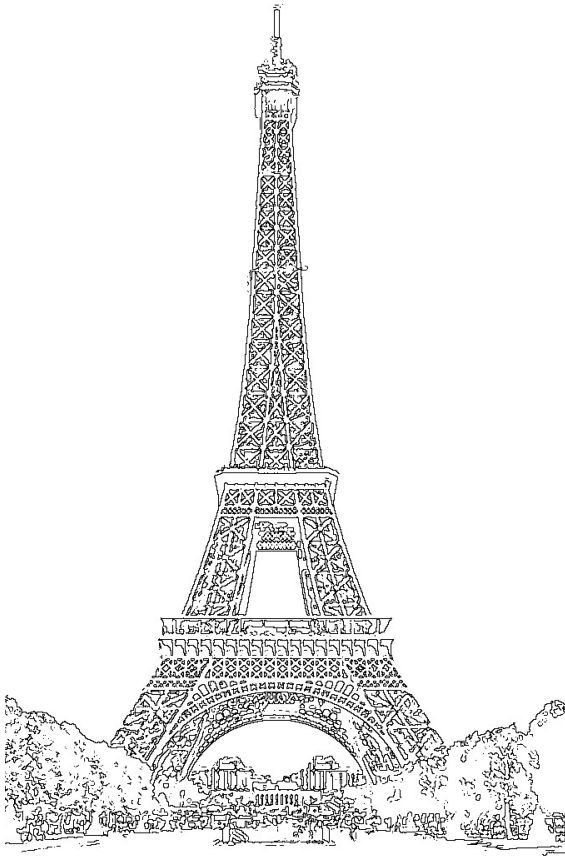


Aplikační oblasti

- Kulturní dědictví



Image Proc. x Comput. Graphics



Digital Image Processing

- Digitization (sampling + quantizing)
- Preprocessing (contrast and brightness changes, denoising, sharpening, ...)
- Image analysis (object detection and recognition, scene understanding)
- Image coding (compression)

Mathematical background

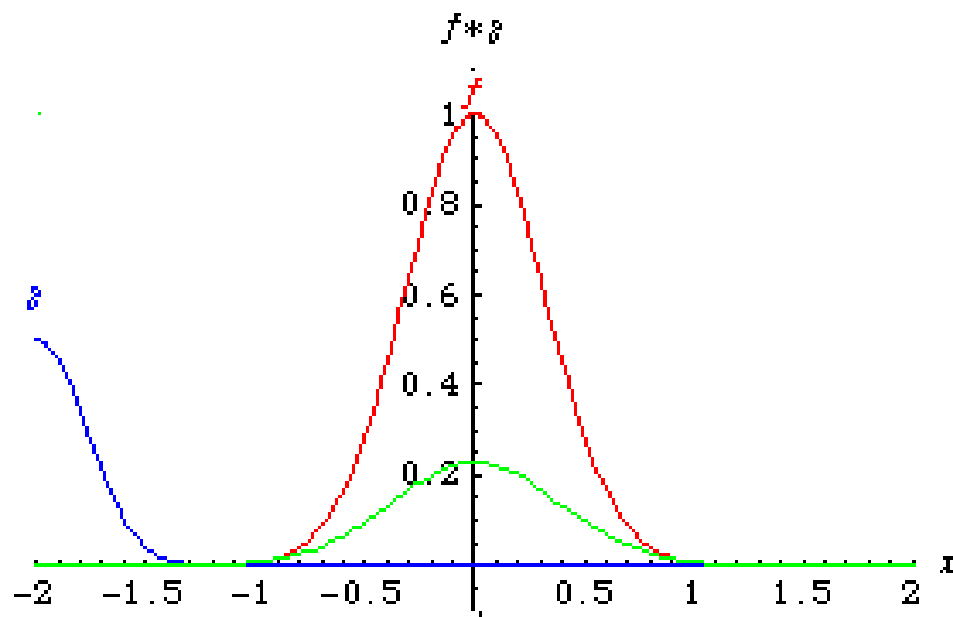
- **Convolution**
- **Fourier Transform**

Convolution

- Definition in continuous domain
- Properties
- Delta function
- Discrete convolution, boundary effect

Convolution

$$* : L_1 \times L_1 \rightarrow L_1 \qquad (f * g)(x) = \int_{-\infty}^{\infty} f(t)g(x - t)dt$$



Basic properties

$$f * g = g * f$$

$$f * (g * h) = (f * g) * h$$

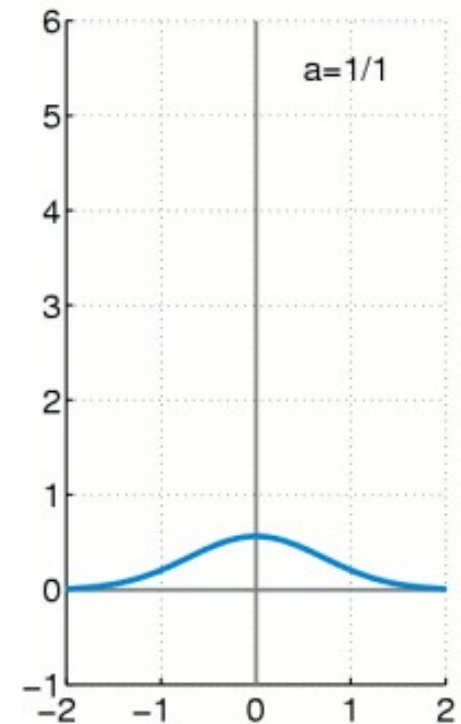
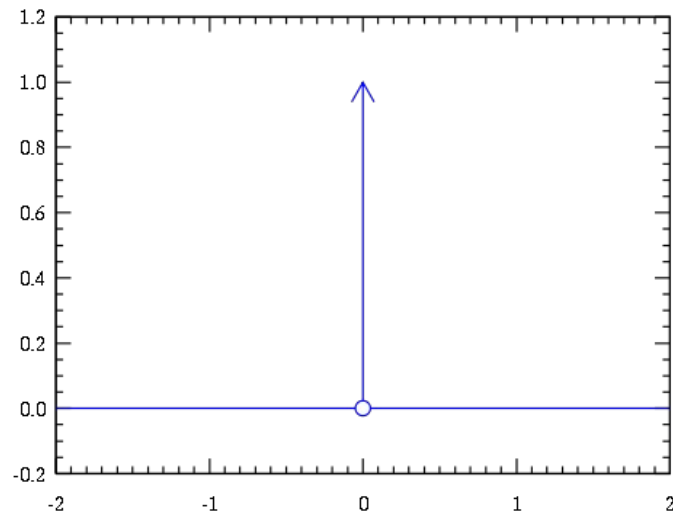
$$a(f * g) = (af) * g = f * (ag)$$

$$f * (g + h) = (f * g) + (f * h)$$

Dirac delta-function

$$\delta(x) = \begin{cases} \infty, & x = 0 \\ 0, & x \neq 0 \end{cases}$$

$$\int \delta(x) dx = 1$$



Delta-function properties

$$\delta(ax) = \frac{\delta(x)}{|a|}$$

$$f * \delta = f$$

$$(f * \delta)(a) = \int_{-\infty}^{\infty} f(x)\delta(a-x)dx = f(a)$$

$$(f * \delta)(x-a) = f(x-a)$$

Cross-correlation

$$(f \star g)(x) = \int_{-\infty}^{\infty} f(t)g(t - x)dt$$

$$f \star g \neq g \star f$$

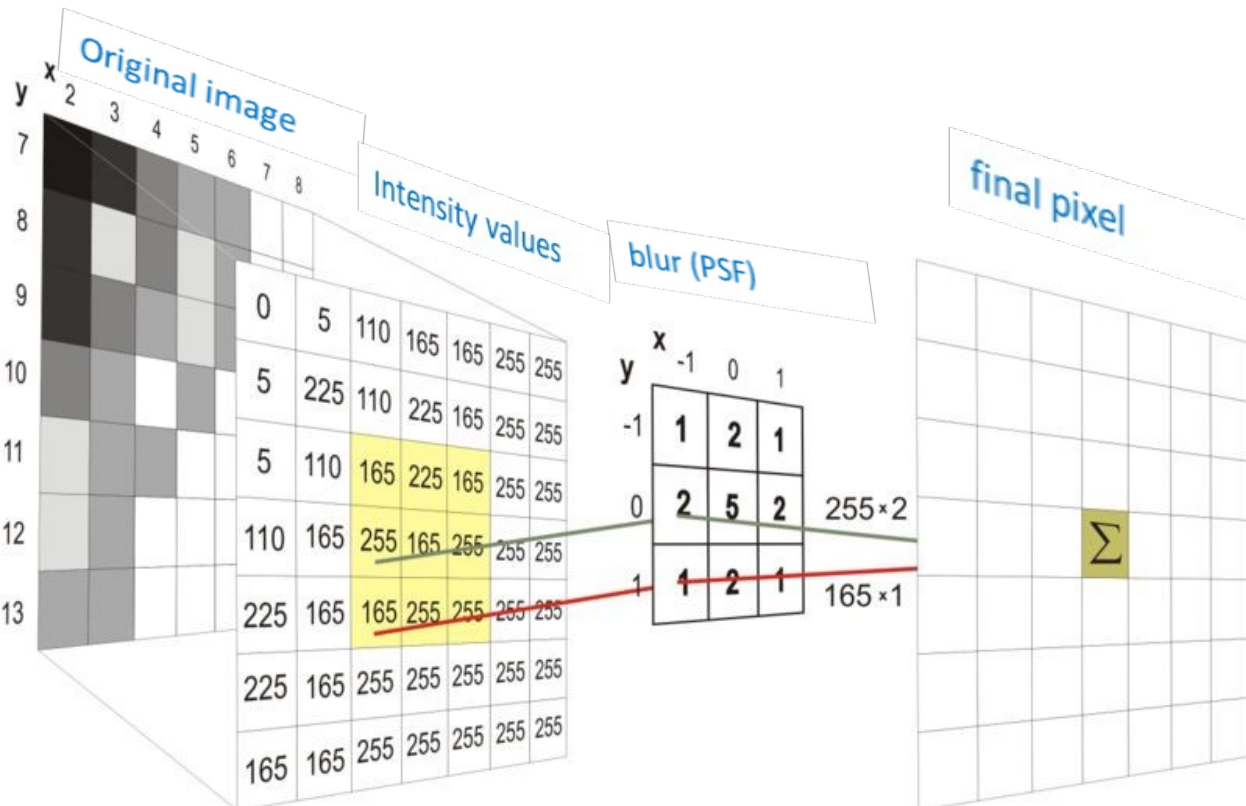
Convolution in 2D

- Definition $* : L_1 \times L_1 \rightarrow L_1$

$$(f * g)(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(s, t) g(x - s, y - t) ds dt$$

Discrete convolution

$$(f * g)(n) = \sum_{m=-\infty}^{\infty} f[m] \cdot g[n - m]$$



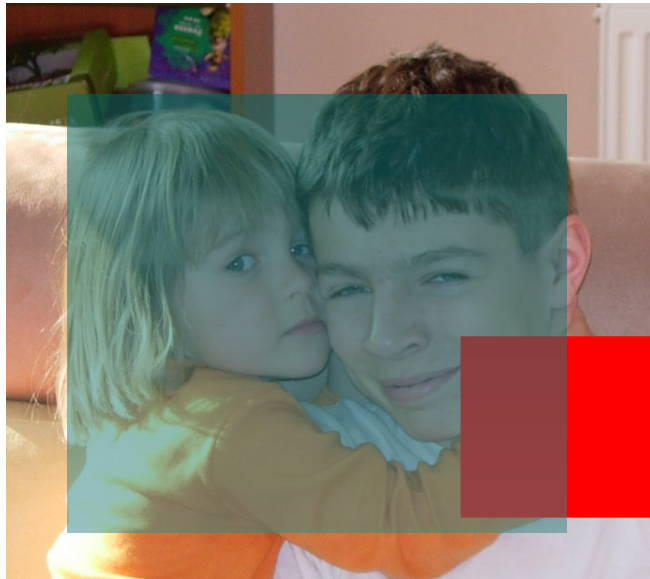
Boundary effect



Convolution
mask







Type
“valid”



Type
“same”



Type
“full”

Boundary effect treatment

- Zero padding
- Mirror extension
- Periodic extension







Computing complexity

- From definition – $O(N^2.M^2)$
- For a separable kernel (rank = 1) – $O(N^2.M)$

$$g(x, y) = g_1(x) \cdot g_2(y)$$

$$(f * g)(x, y) = \int_{-\infty}^{\infty} g_1(x-s) \left(\int_{-\infty}^{\infty} f(s, t) g_2(y-t) dt \right) ds$$

- Hardware acceleration (DSP, graphic cards, ...)

Fourier transform



Recalling Fourier series

$\{\phi_k\}$ – ON basis of a Hilbert space $\mathcal{H}$, $f \in \mathcal{H}$

$c_k = (f, \phi_k)$ – Fourier coefficients

$$f = \sum_{i=1}^{\infty} c_k \phi_k$$

Fourier series in $L_2(-\pi, \pi)$

$$\{\phi_k\} = \{1, \sin nx, \cos mx\}$$

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx)$$

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos kx \, dx$$

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin kx \, dx$$

Fourier transform

$$\{\phi_u(x)\} = \{e^{-2\pi i(ux)}\}$$

$$F(u) = \int_{\mathbb{R}} f(x)e^{-2\pi iux} \mathrm{d}x$$

$$f(x) = \int_{\mathbb{R}} F(u)e^{2\pi iux} \mathrm{d}u$$

Properties of the FT

If $f \in L_1$ then FT exists

Generally $F \notin L_1$

FT is "one-to-one" mapping: $f \longleftrightarrow F$

$$F = \mathcal{R}(F) + i\mathcal{I}(F) = |F| \cdot e^{i \cdot ph(F)}$$

Properties of the FT

- **linearity**

$$\mathcal{F}[af(x) + bg(x)] = a\mathcal{F}[f(x)] + b\mathcal{F}[g(x)] = aF(k) + bG(k).$$

- **convolution**

convolution theorem

$$\mathcal{F}[f]\mathcal{F}[g] = \mathcal{F}[f * g]$$

- **shift**

shift theorem

$$\mathcal{F}_x[f(x - x_0)](k) = e^{-2\pi i k x_0} F(k).$$

- **rotation**

$$F(R(f)) = R(F(f))$$

- **scaling**

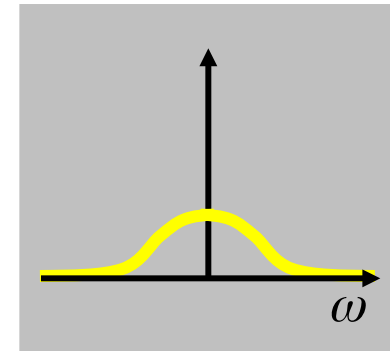
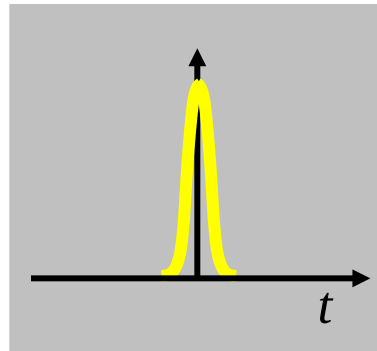
similarity theorem

$$\mathcal{F}_x[f(ax)](k) = |a|^{-1} F\left(\frac{k}{a}\right).$$

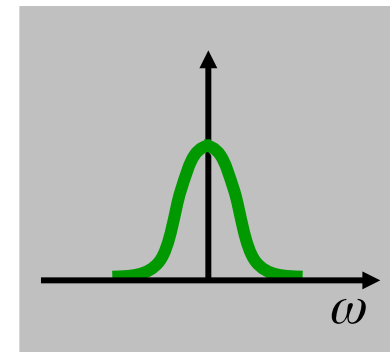
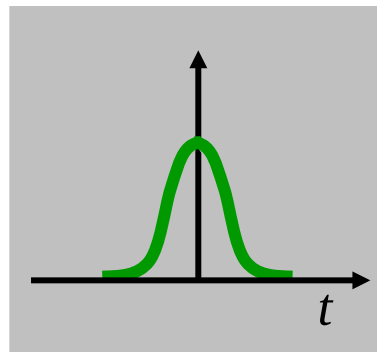
scaling
similarity theorem

$$\mathcal{F}_x[f(ax)](k) = |a|^{-1} F\left(\frac{k}{a}\right).$$

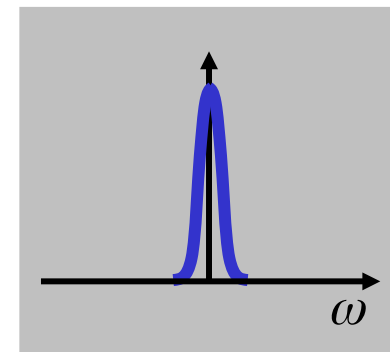
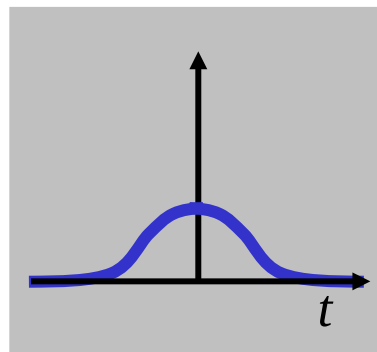
Short pulse



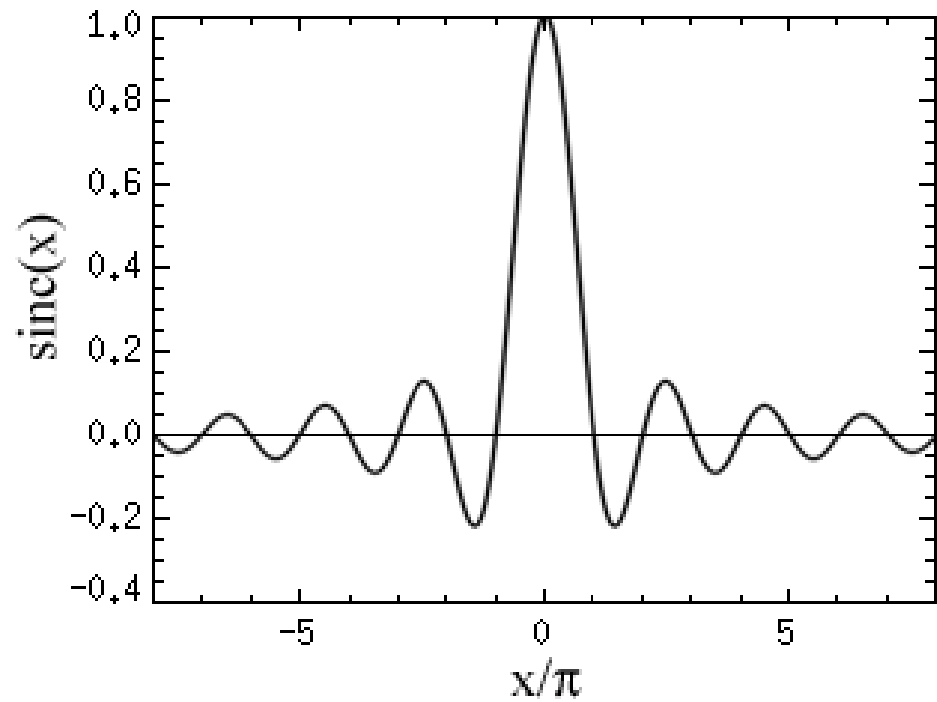
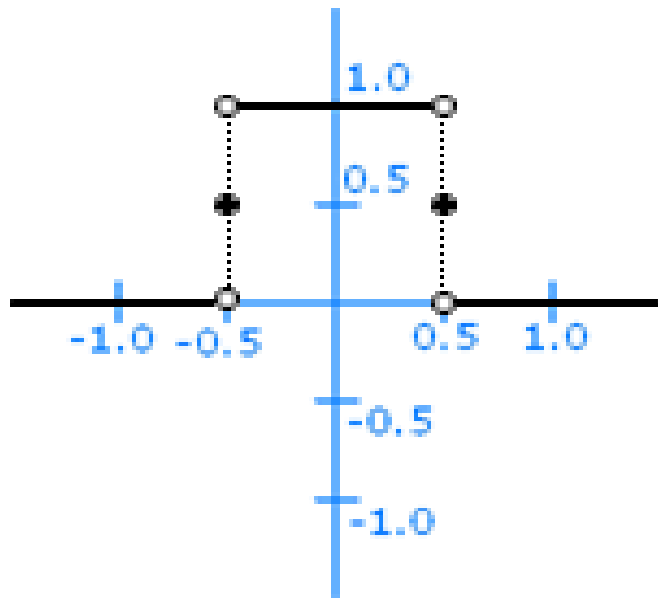
Medium-length pulse



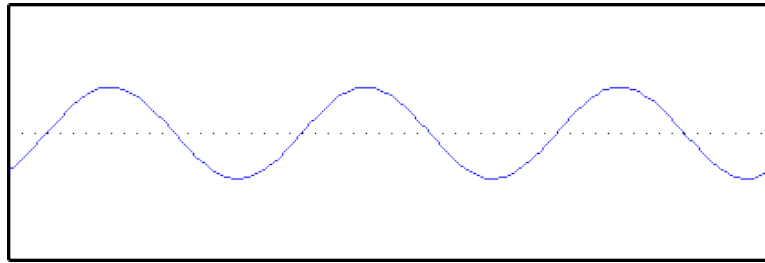
Long pulse



Rectangular pulse



Harmonic wave



FT is a delta-function properly shifted

Discrete Fourier Transform

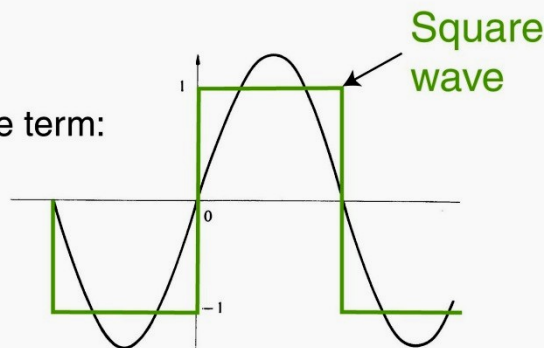
$$F_n \equiv \sum_{k=0}^{N-1} f_k e^{-2\pi i n k / N}.$$

$$f_k = \frac{1}{N} \sum_{n=0}^{N-1} F_n e^{2\pi i n k / N}.$$

$$\sin(\omega t)$$



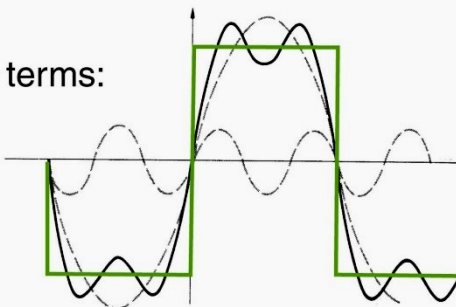
One term:



$$\sin(3\omega t)$$



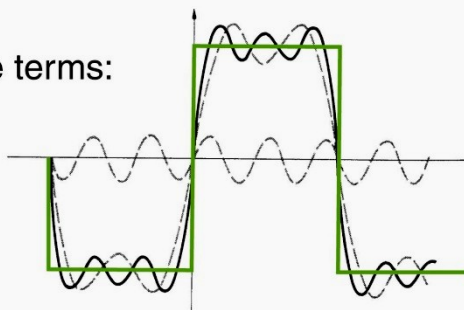
Two terms:



$$\sin(5\omega t)$$

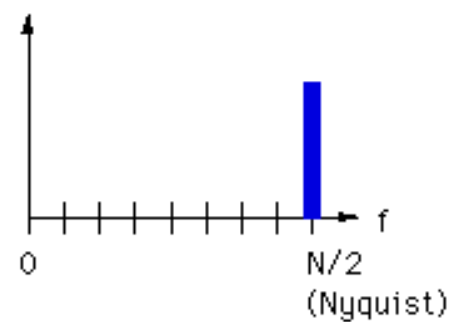
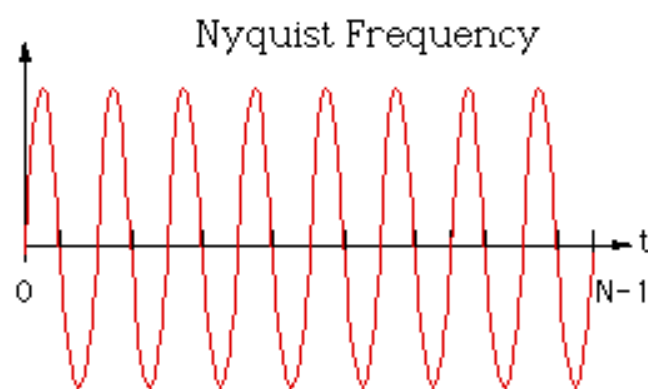
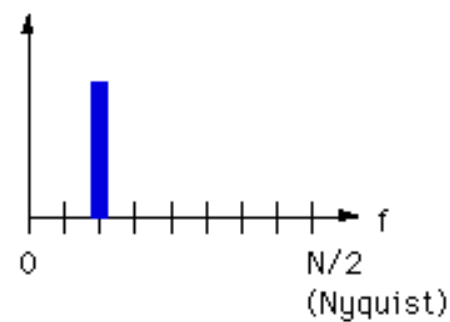
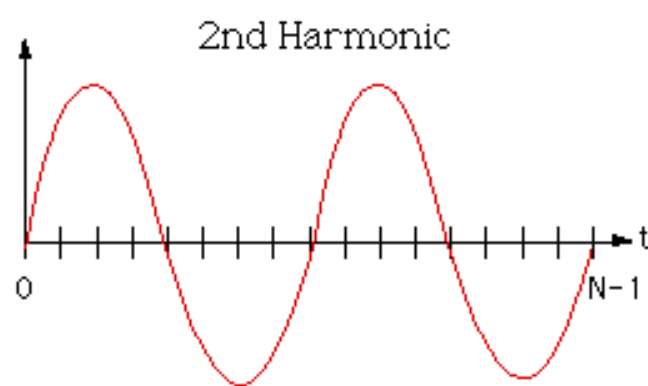
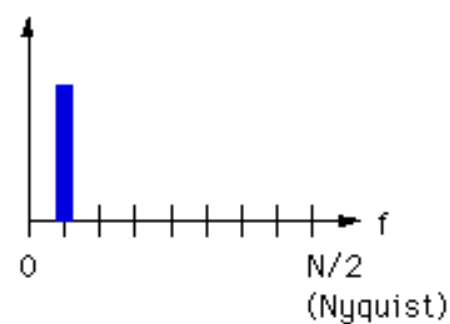
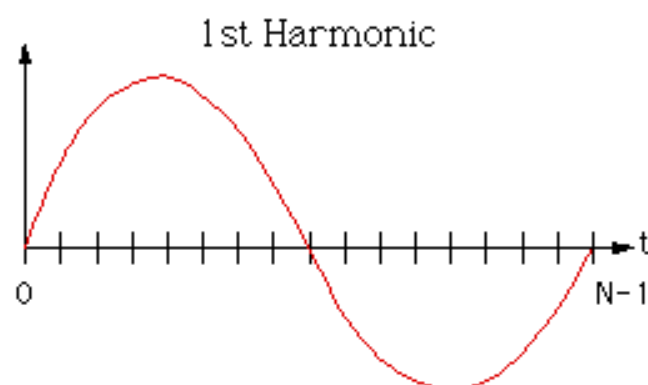


Three terms:



Discrete Fourier Transform

- What are the frequencies in DFT?
- Which frequency is the lowest/highest one?
- Formally DFT exists for any $n \rightarrow$ periodicity. Only N samples are independent.



DFT calculation

$$F_n \equiv \sum_{k=0}^{N-1} f_k e^{-2\pi i n k / N}$$

Directly - $O(N^2)$

FFT - $O(N \log N)$

(Cooley and Tookey 1965, Gauss 1866)

Compare to the complexity of convolution.

2D Fourier transform

$$\{\phi_{uv}(x, y)\} = \{e^{-2\pi i(ux+vy)}\}$$

$$F(u, v) = \int \int_{\mathbb{R}^2} f(x, y) e^{-2\pi i(ux+vy)} dx dy$$

$$f(x, y) = \int \int_{\mathbb{R}^2} F(u, v) e^{2\pi i(ux+vy)} du dv$$

2D Fourier transform is separable

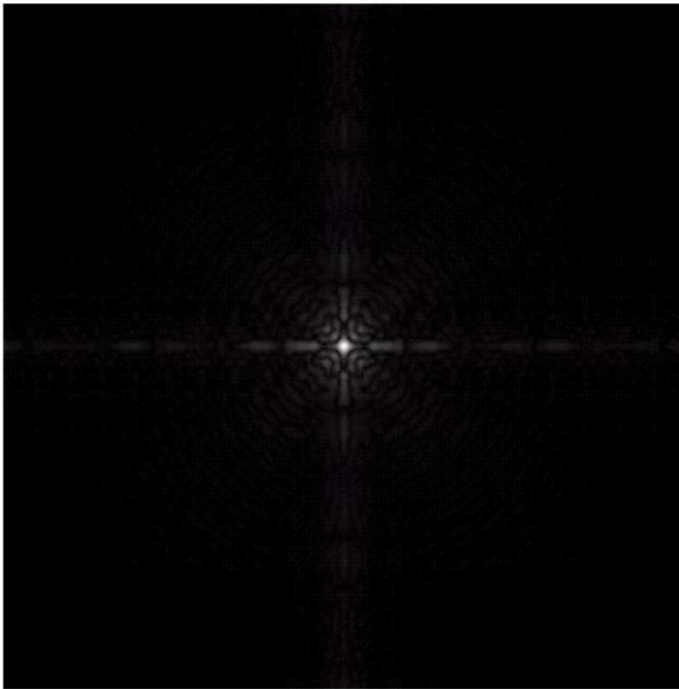
Visualisation



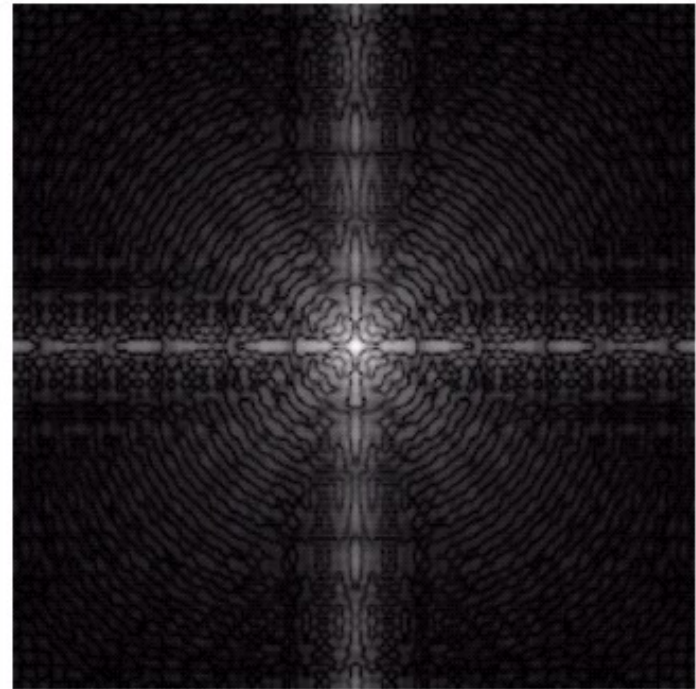
Amplitude

Phase

Visualisation

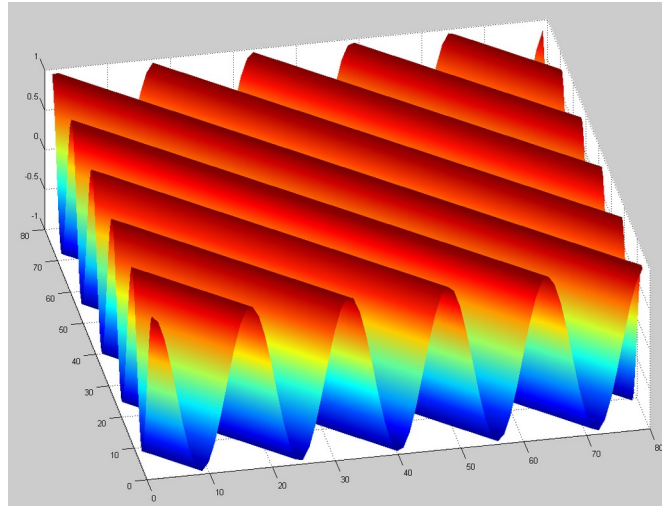


ampl



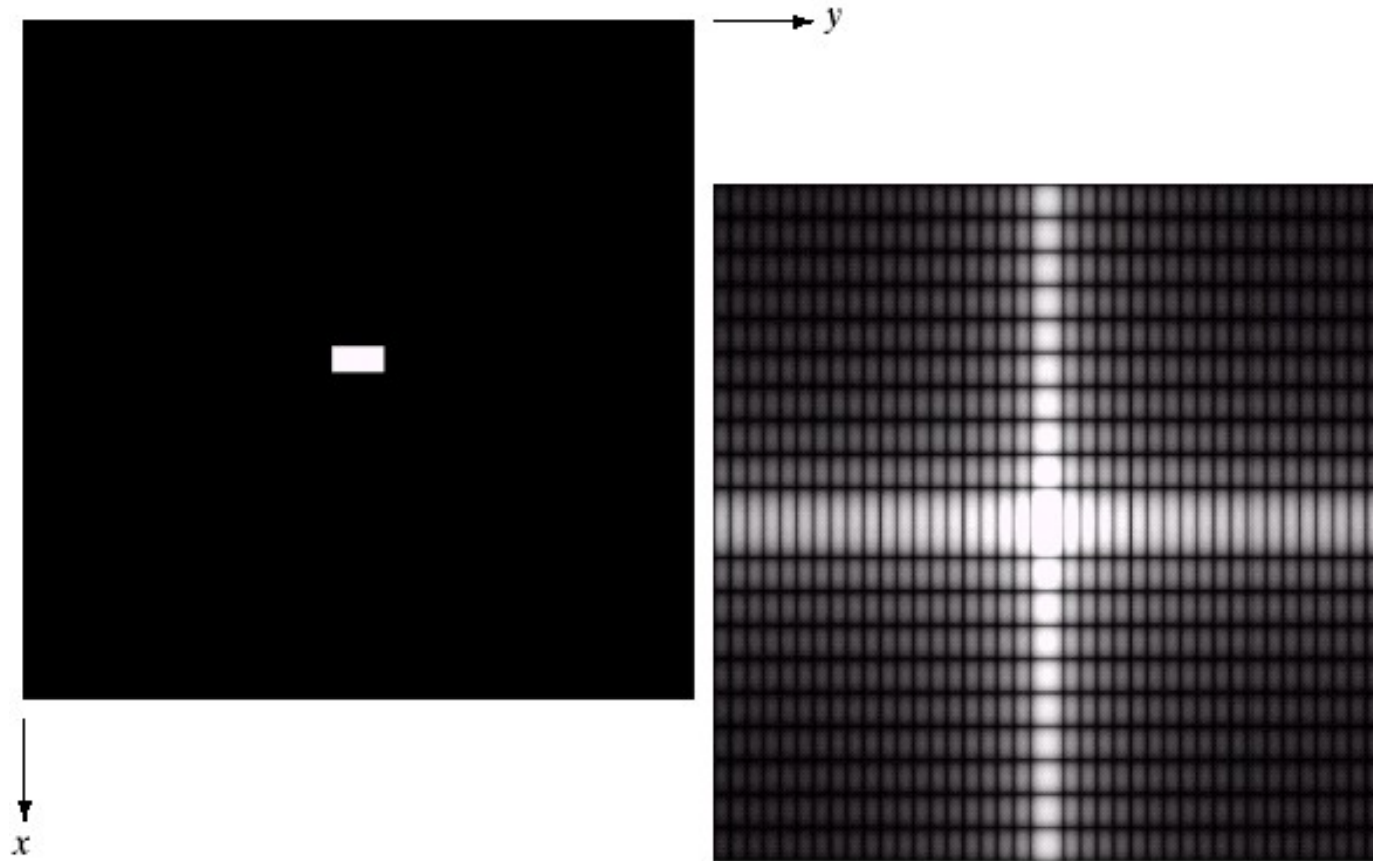
$\log(\text{ampl} + 1)$

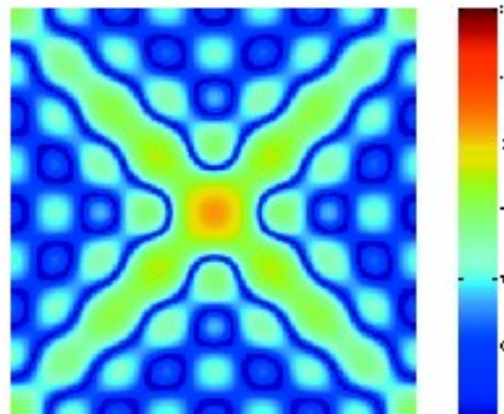
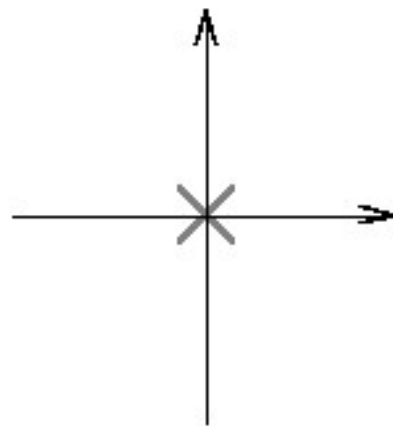
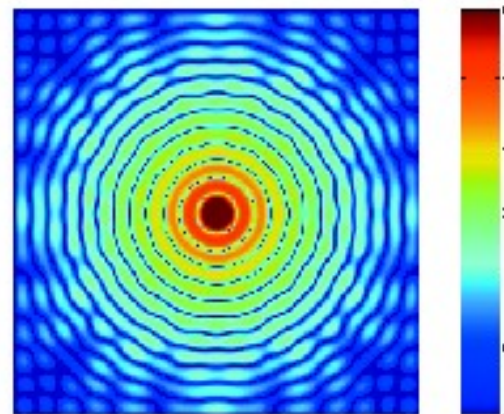
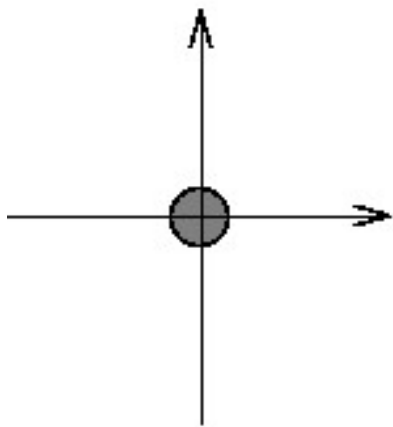
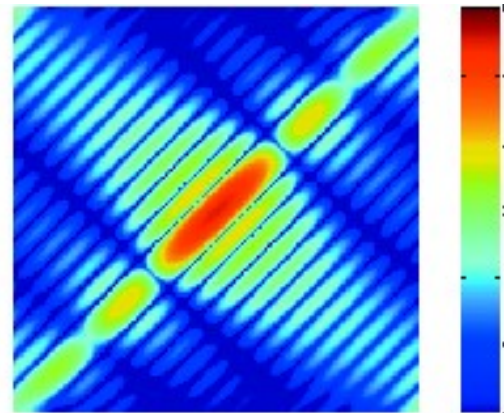
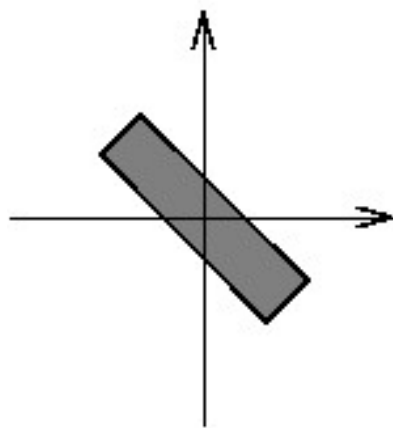
Basis functions of 2D FT

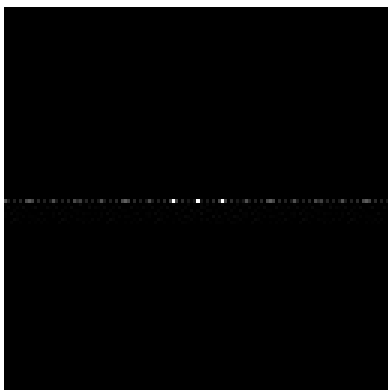
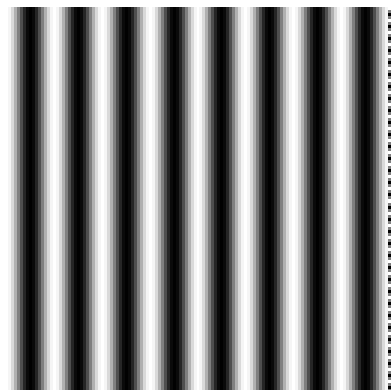
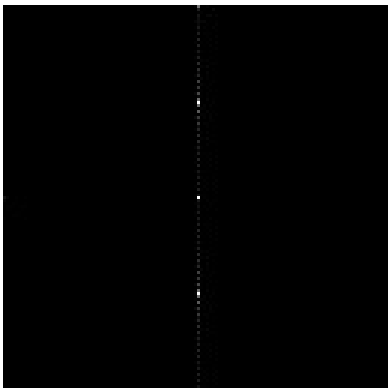
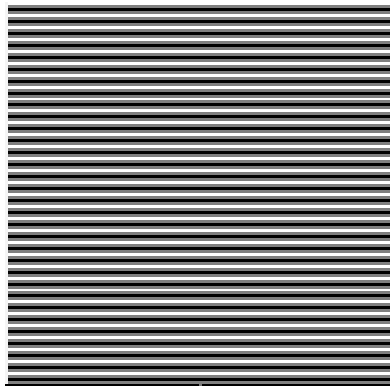


Real part, $u=v$

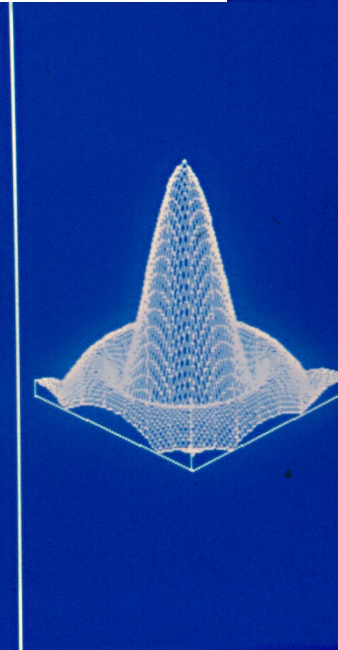
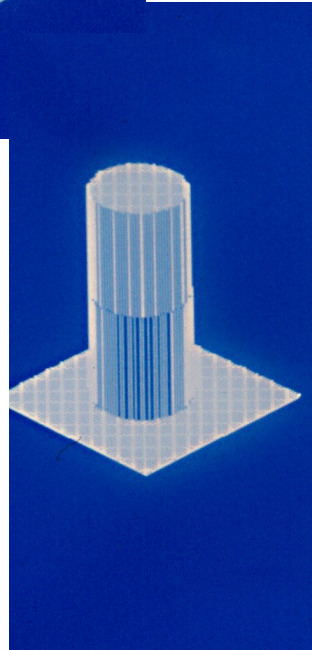
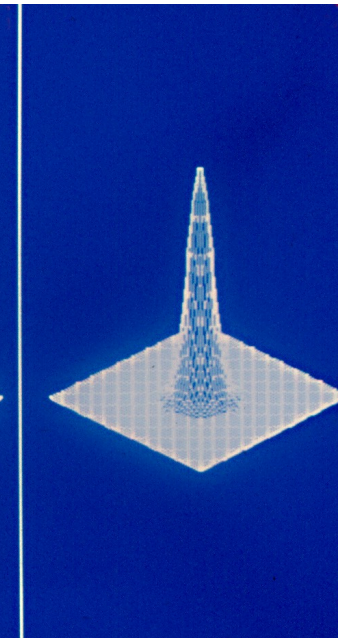
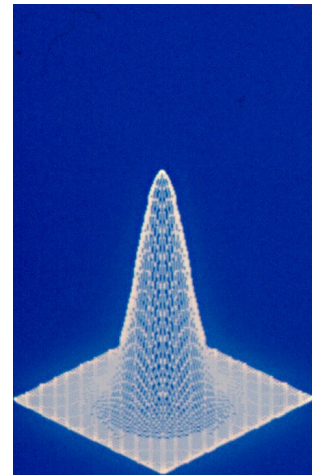
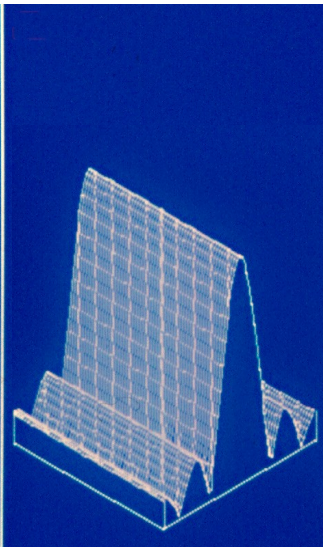
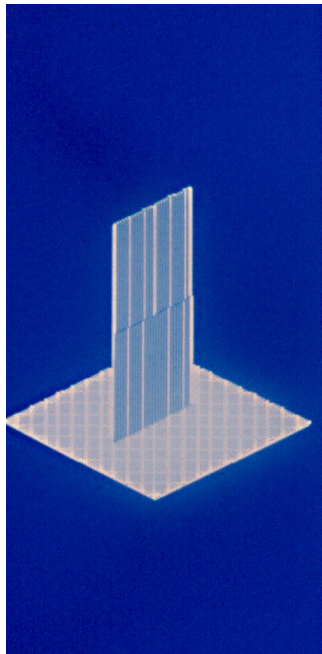
Rectangular pulse in 2D







Other important functions



Discrete convolution theorem



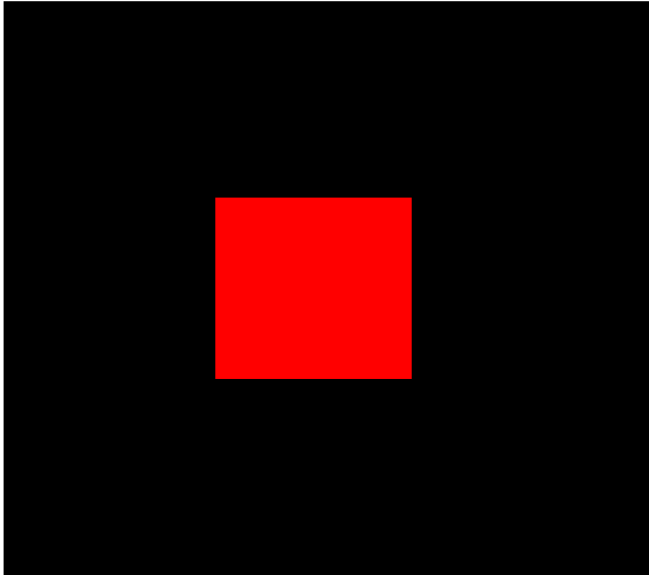
Convolution
mask



... holds for periodic convolution only!



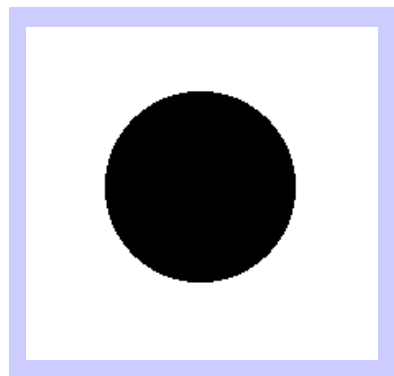
Discrete convolution via FT



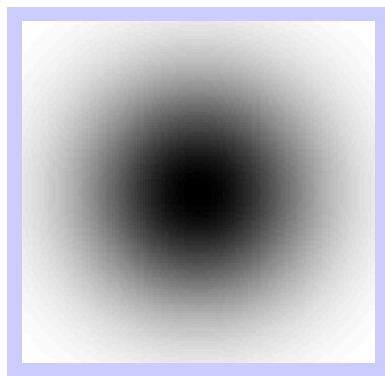
Zero padding to the same size
DFT calculation of both
Multiplication of the spectra
Inverse DFT

Filtering in the Fourier domain

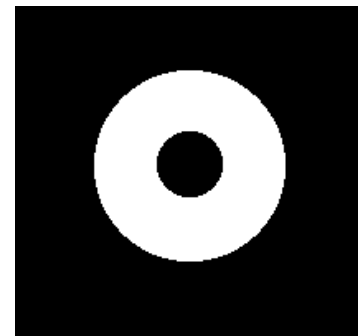
$$\mathcal{F}[f * g] = \mathcal{F}[f]\mathcal{F}[g]$$



high pass

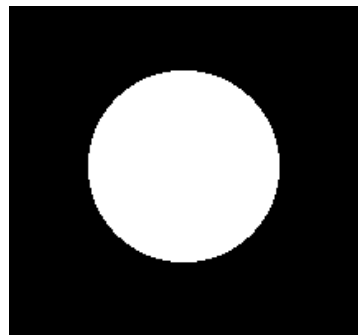


Gaussian high pass

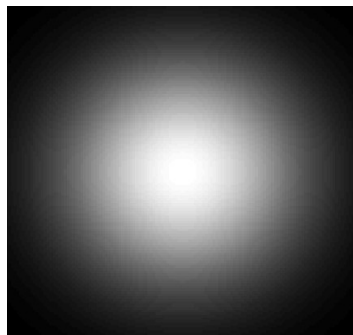


band pass

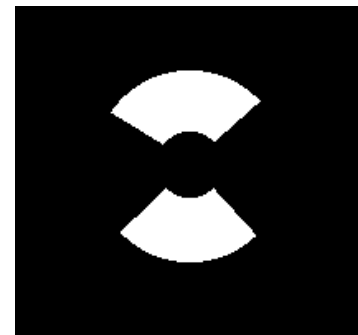
low pass

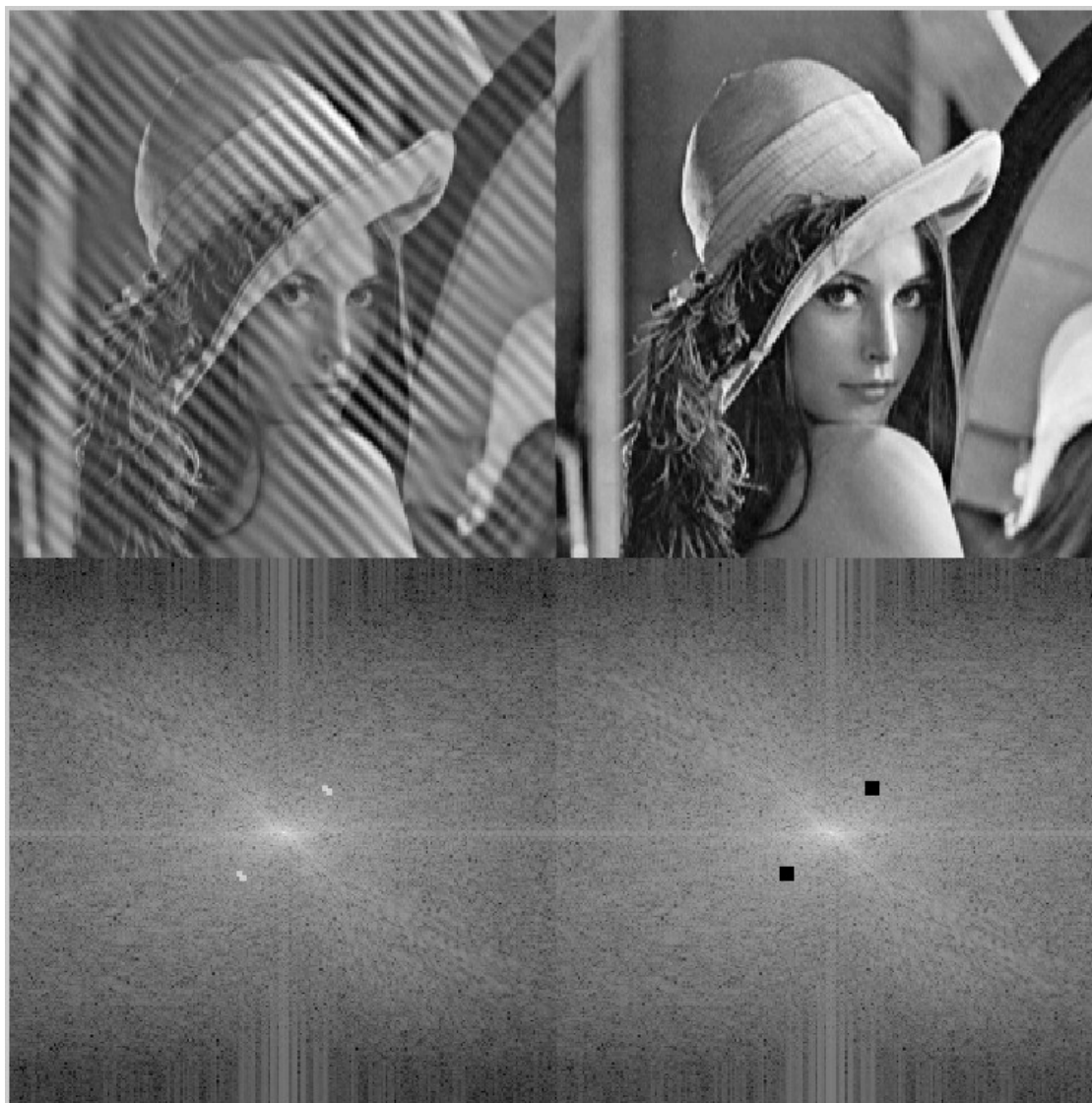


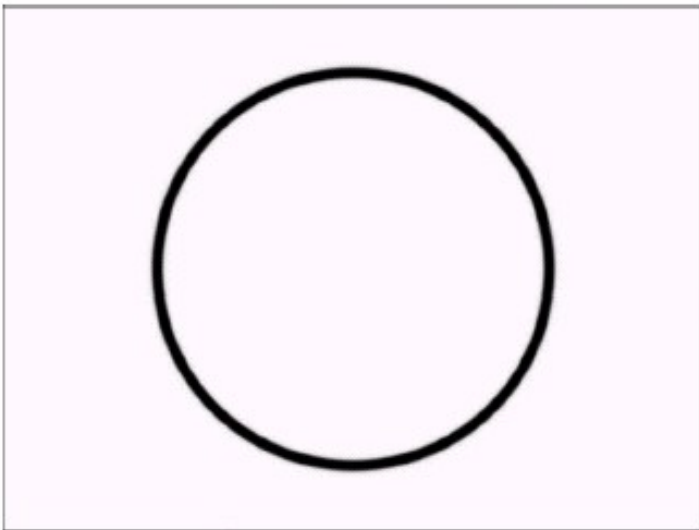
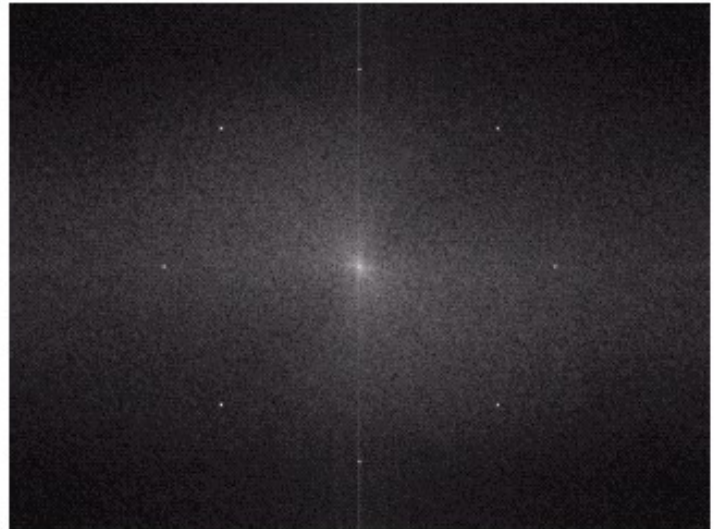
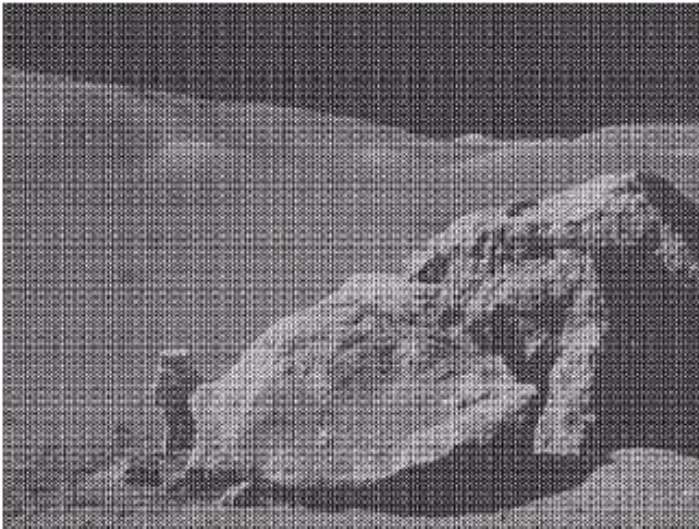
Gaussian low pass



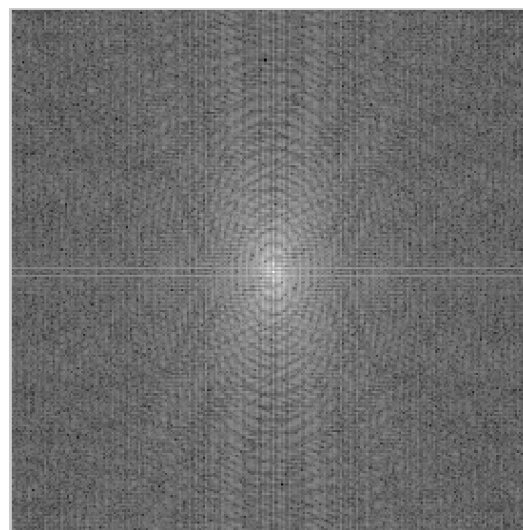
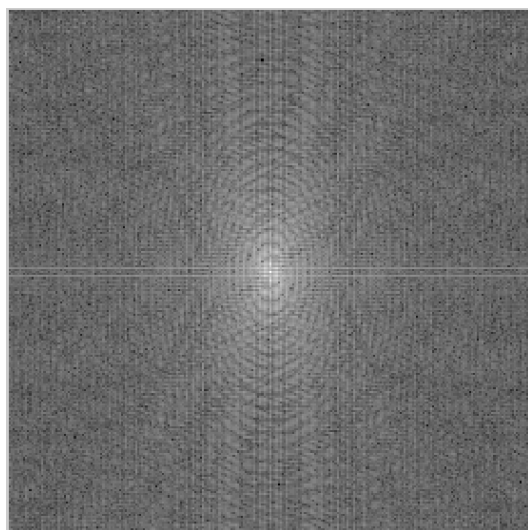
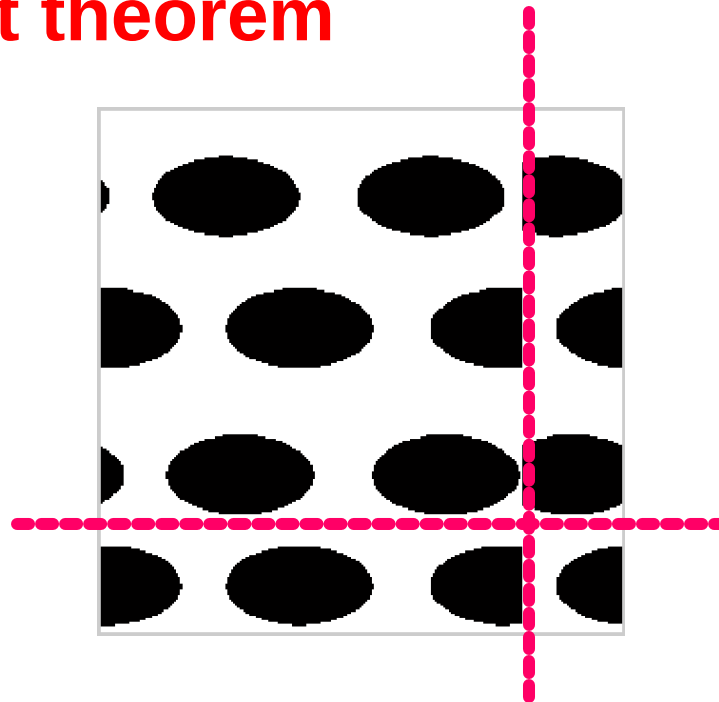
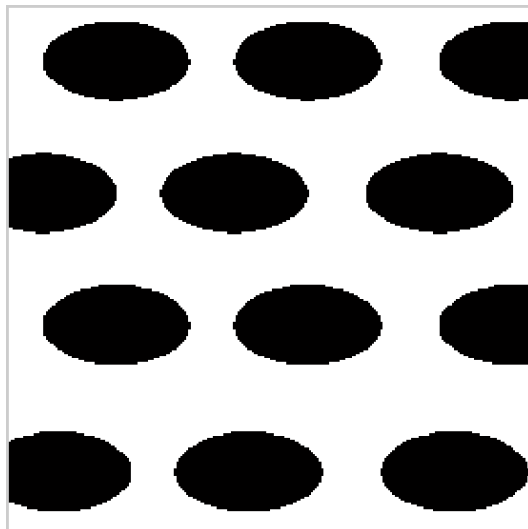
directional

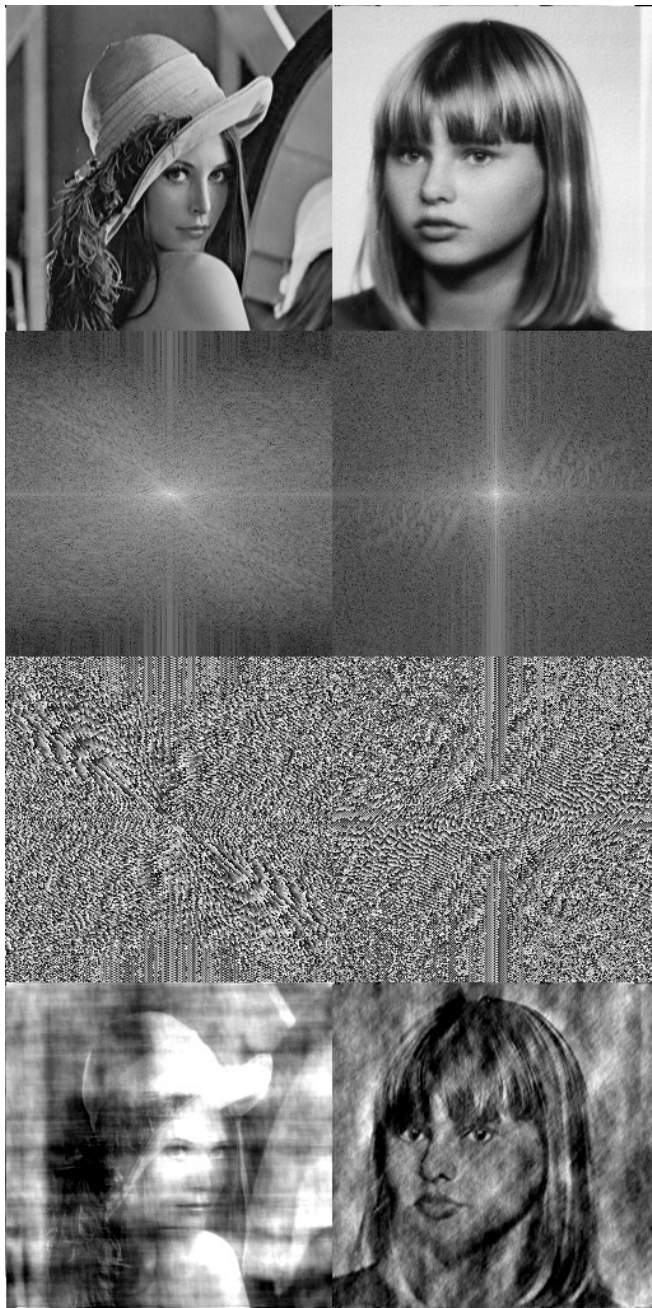






Discrete shift theorem





What is more
important?

The phase!

Image whitening

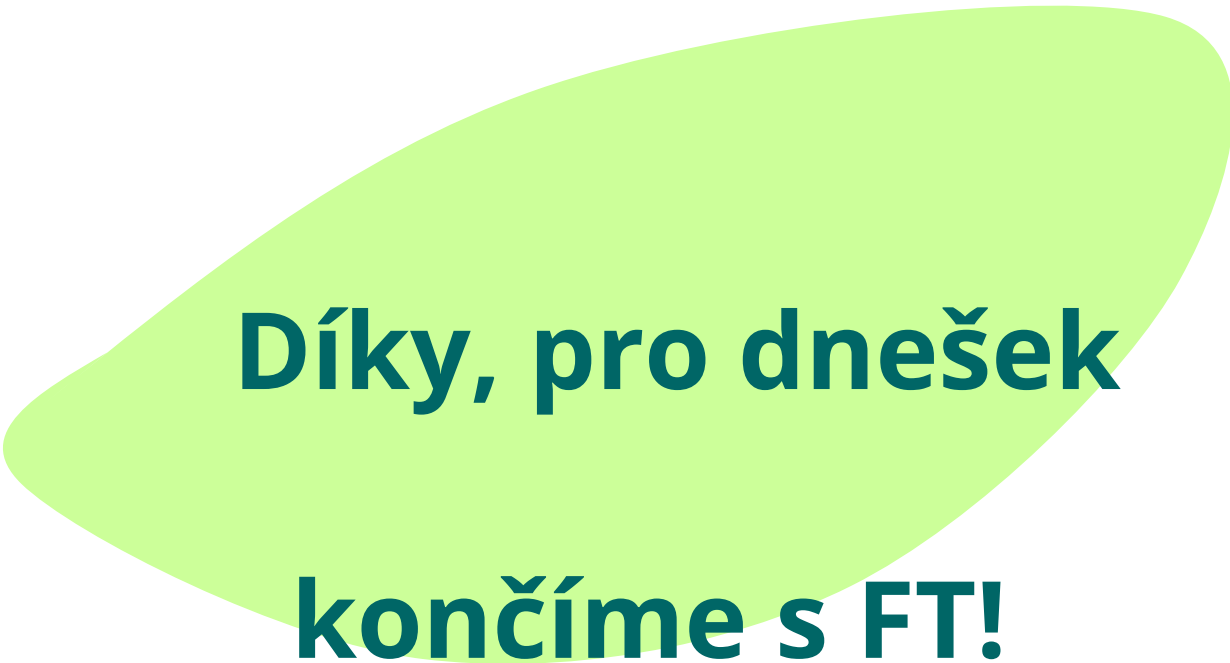
$$w(x, y) = \mathcal{F}^{-1} \frac{F(u, v)}{|F(u, v)|}$$



original image



“whitened” image



**Díky, pro dnešek
končíme s FT!**

Nějaké otázky ?